Foundations of Operations Research

Edoardo Amaldi (edoardo.amaldi@polimi.it)

Dipartimento di Elettronica, Informazione e Bioingegneria – Politecnico di Milano

2025/26

Chapter 3: Linear Programming (LP)

Definition 1

A Linear Programming (LP) problem is an optimization problem:

$$\begin{array}{ll} \min & f(\underline{x}) \\ \text{s.t.} & \underline{x} \in X \subseteq \mathbb{R}^n \end{array} \tag{1}$$

where:

- the objective function $f : X \to \mathbb{R}$ is linear;
- the feasible region $X = \{\underline{x} \in \mathbb{R}^n : g_i(\underline{x}\ )\ r_i\ 0, i \in \{1, \ldots, m\}, r_i \in \{=, \geq, \leq\}\}$ with $g_i : \mathbb{R}^n \to \mathbb{R}$ linear, for $i = 1, \ldots, m$.

Definition 2

$\underline{x}^* \in \mathbb{R}^n$ is an optimal solution of (1) if $f(\underline{x}^*) \leq f(\underline{x})$, $\forall \underline{x} \in X$.

A variety of decision-making problems can be formulated/approximated as LPs.

Often involve the optimal allocation of limited resources to different activities.

General form:

$$\begin{array}{ll} \min & z = c_1 x_1 + \cdots + c_n x_n \\ \text{s. t.} & a_{11} x_1 + \cdots + a_{1n} x_n \ (\geq, =, \leq) \ b_1 \\ & \vdots \\ & a_{m1} x_1 + \cdots + a_{mn} x_n \ (\geq, =, \leq) \ b_m \\ & x_1, \ldots, x_n \geq 0 \end{array}$$

Matrix notation

$$\begin{array}{ll} \min & z = \underline{c}^T \underline{x} \\ \text{s. t.} & A\underline{x} \geq \underline{b} \\ & \underline{x} \geq 0 \end{array} \quad ,$$

$$\begin{array}{ll} \min & z = [c_1 \ldots c_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ \text{s. t.} & \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \geq \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \\ & \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \geq \underline{0} \end{array}$$

Historical sketch

Jean B. Joseph Fourier
(1786–1830)

L. V. Kantorovitch
(1912–1986)

G. B. Dantzig
(1914–2005)

- 1825/26: Fourier presents a method to solve systems of linear inequalities and discusses LPs with 2-3 variables.
- 1939: Kantorovitch lays the foundations of LP and study the transportation problem (Nobel prize, 1975).
- 1947: Dantzig independently proposes LP and invents the Simplex algorithm.

Example 1: Diet problem

Given

n aliments $j = 1, \dots, n$

m nutrients (basic substances) $i = 1, \dots, m$

a_{ij} amount of i-th nutrient contained in one unit of the j-th aliment

b_i daily requirement of the i-th nutrient

c_j cost of a unit of j-th aliment,

determine a diet that minimizes the total cost while satisfying all daily requirements.

Model:

Decision variables: x_j = amount of j-th aliment in the diet, with $j = 1, \dots, n$

$$\begin{aligned} \min \quad & z = \sum_{j=1}^{n} c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \geq b_i, \quad i = 1, \dots, m \\ & x_j \geq 0, \; j = 1, \dots, n \end{aligned}$$

Example 2: Transportation problem (single product)

Given

m production plants $i = 1, \ldots, m$

n clients $j = 1, \ldots, n$

c_{ij} unit transportation cost from plant i to client j

p_i maximum supply (production capacity) of plant i

d_j demand of client j

q_{ij} maximum amount transportable from plant i to client j

determine a transportation plan that minimizes the total costs while respecting plant capacities and client demands

Assumption: $\sum_{i=1}^{m} p_i \geq \sum_{j=1}^{n} d_j$

Model:

Decision variables: x_{ij} = amount of product transported from i to j, with $1 \leq i \leq m$ and $1 \leq j \leq n$

$$
\begin{aligned}
\min \quad & z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} \\
\text{s.t.} \quad & \sum_{j=1}^{n} x_{ij} \leq p_i, \quad i = 1, \ldots, m && \text{(plant capacity)} \\
& \sum_{i=1}^{m} x_{ij} \geq d_j, \quad j = 1, \ldots, n && \text{(client demand)} \\
& 0 \leq x_{ij} \leq q_{ij}, \; i = 1, \ldots, m, \; j = 1, \ldots, n && \text{(transport. capacity)}
\end{aligned}
$$

Example 3: Production planning problem

Given

m resources $i = 1, \dots, m$

n products $j = 1, \dots, n$ which compete for resources

c_j profit (selling price-cost) per unit of j-th product

a_{ij} amount of i-th resource needed to produce one unit of j-th product

b_i maximum available amount of i-th resource

determine a production plan that maximizes the total profit given the available resources.

Model: Decision variables: x_j = amount of j-th product, with $j = 1, \dots, n$

$$\begin{aligned} \max \quad & z = \sum_{j=1}^{n} c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \leq b_i, \quad i = 1, \dots, m \\ & x_j \geq 0, \; j = 1, \dots, n \end{aligned}$$

Assumptions of LP models

1. Linearity of the objective function and constraints.

   Proportionality: Contribution of each variable = constant $\times$ variable
   Drawback: does not account for economies of scale.

   Additivity: Contribution of all variables = sum of single contributions
   Drawback: Competing products $\Rightarrow$ profits are not independent.

2. Divisibility: The variables can take fractional (rational) values.

3. Parameters are considered as constants which can be estimated with a sufficient degree of accuracy.

   More complex mathematical programs are needed to account for uncertainty.

LP "sensitivity analysis" allows to evaluate how sensitive an optimal solution is with respect to small changes in the parameter values (see end of Chapter 3).

3.1 Equivalent forms

General form:

$$
\begin{array}{lll}
\min(\max) & z = \underline{c}^T \underline{x} & \\
\text{s. t.} & A_1 \underline{x} \geq \underline{b_1} & \text{inequalities} \\
 & A_2 \underline{x} \leq \underline{b_2} & \text{inequalities} \\
 & A_3 \underline{x} = \underline{b_3} & \text{equations} \\
 & x_j \geq 0, \quad \text{for } j \in J \subseteq \{1, \ldots, n\} & \\
 & x_j \text{ free}, \quad \text{for } j \in \{1, \ldots, n\} \backslash J &
\end{array}
$$

Definition 3 (Standard form)

$$
\begin{array}{lll}
\min & z = \underline{c}^T \underline{x} & \\
\text{s. t.} & A\underline{x} = \underline{b} & \text{only equality constraints and} \\
 & \underline{x} \geq \underline{0} & \text{all nonnegative variables}
\end{array}
$$

- The two forms are equivalent.
- Simple transformation rules allow to pass from one form to the other form.

Warning: The transformation may involve adding/deleting variables and/or constraints.

Transformation rules

- $\max \underline{c}^T \underline{x} = -\min(-\underline{c}^T \underline{x})$
- $\underline{a}^T \underline{x} \leq b \Rightarrow \begin{cases} \underline{a}^T \underline{x} + s = b \\ s \geq 0 \end{cases}$ where s is a slack variable.
- $\underline{a}^T \underline{x} \geq b \Rightarrow \begin{cases} \underline{a}^T \underline{x} - s = b \\ s \geq 0 \end{cases}$ where s is a surplus variable.
- x_j unrestricted in sign $\Rightarrow \begin{cases} x_j = x_j^+ - x_j^- \\ x_j^+, x_j^- \geq 0 \end{cases}$.

  After substituting x_j with $x_j^+ - x_j^-$, we delete x_j from the problem.

Example

General form:

$$\begin{array}{ll} \max & f(x_1, x_2) = 2x_1 - 3x_2 \\ \text{s. t.} & 4x_1 - 7x_2 \leq 5 \\ & 6x_1 - 2x_2 \geq 4 \\ & x_1 \geq 0, \quad x_2 \in \mathbb{R} \end{array}$$

1. $x_2 = x_3 - x_4$, $x_3, x_4 \geq 0$:

$$\begin{array}{ll} \max & 2x_1 - 3x_3 + 3x_4 \\ \text{s. t.} & 4x_1 - 7x_3 + 7x_4 \leq 5 \\ & 6x_1 - 2x_3 + 2x_4 \geq 4 \\ & x_1, x_3, x_4 \geq 0 \end{array}$$

2. Introduce slack and surplus variables x_5 and x_6:

$$\begin{array}{ll} \max & 2x_1 - 3x_3 + 3x_4 \\ \text{s. t.} & 4x_1 - 7x_3 + 7x_4 + x_5 = 5 \\ & 6x_1 - 2x_3 + 2x_4 - x_6 = 4 \\ & x_1, x_3, x_4, x_5, x_6 \geq 0 \end{array}$$

3. Change the objective function sign:

$$\begin{array}{ll} \min & -2x_1 + 3x_3 - 3x_4 \\ \text{s. t.} & 4x_1 - 7x_3 + 7x_4 + x_5 = 5 \\ & 6x_1 - 2x_3 + 2x_4 - x_6 = 4 \\ & x_1, x_3, x_4, x_5, x_6 \geq 0 \end{array}$$

Other straightforward transformations

- $a^T x \leq b \Leftrightarrow -a^T x \geq -b$
- $a^T x \geq b \Leftrightarrow -a^T x \leq -b$
- $a^T x = b \Leftrightarrow \begin{cases} a^T x \geq b \\ a^T x \leq b \end{cases} \Leftrightarrow \begin{cases} a^T x \geq b \\ -a^T x \geq -b \end{cases}$

3.2 Geometry of Linear Programming

Example Capital budgeting

Capital of 10 000€ and two possible investments A and B with, respectively, 4% and 6% expected return.

Determine a portfolio that maximizes the total expected return, while respecting the diversification constraints:

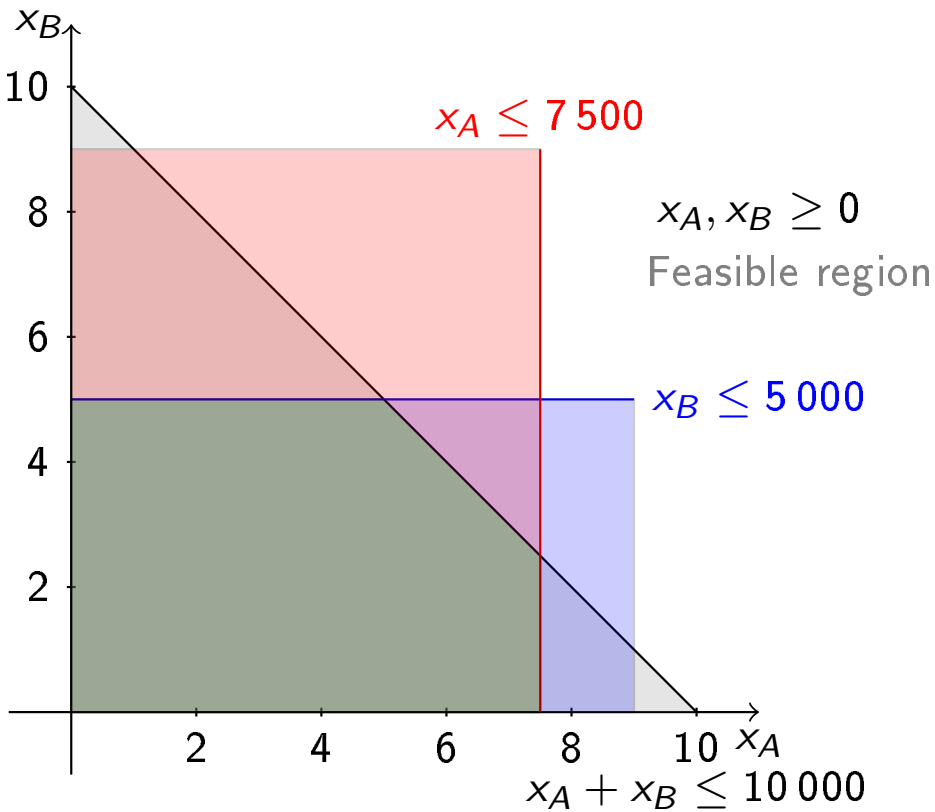
- at most 75% of the capital is invested in A,
- at most 50% of the capital is invested in B.

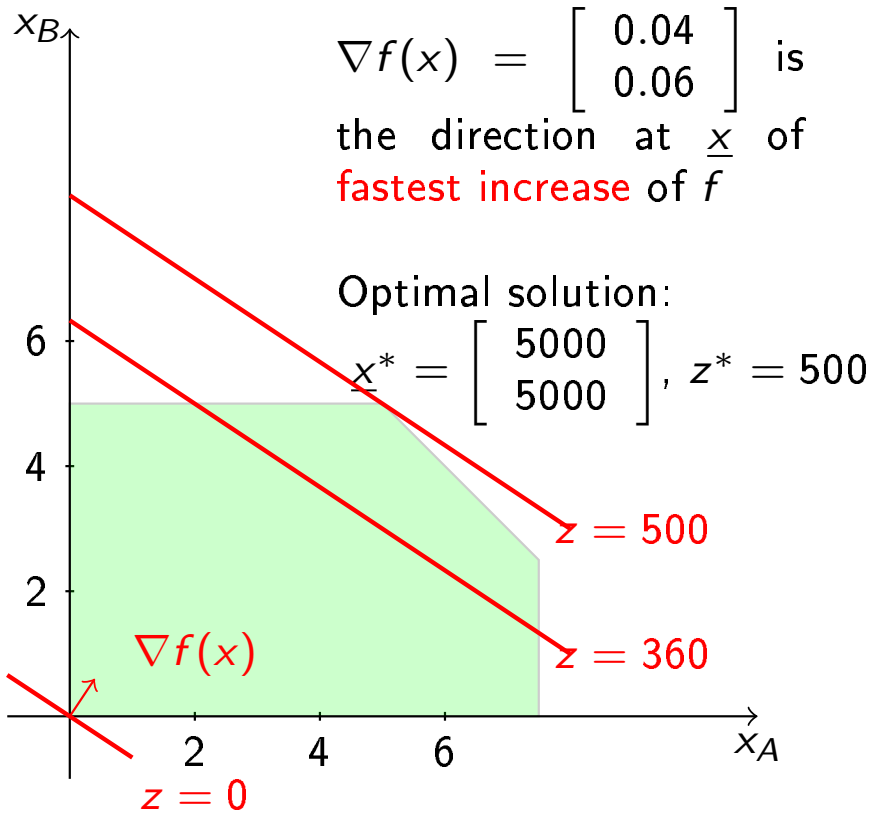
Model:

- x_A = amount invested in A
- x_B = amount invested in B

$$
\begin{array}{llrrcl}
\max & z = 0.04x_A & +0.06x_B & & & \\
\text{s.t.} & x_A & +x_B & \leq & 10\,000 \\
 & x_A & & \leq & 0.75 \cdot 10\,000 \\
 & & x_B & \leq & 0.50 \cdot 10\,000 \\
 & x_A, & x_B & \geq & 0
\end{array}
$$

3.2.1 Graphical solution





Definition 4

A level curve of value z of a function f is the set of points in $\mathbb{R}^n$ where f is constant and takes value z.

The level curves of a LP are lines: $0.04x_A + 0.06x_B = z$ (z is a constant)

3.2.2 Vertices of the feasible region

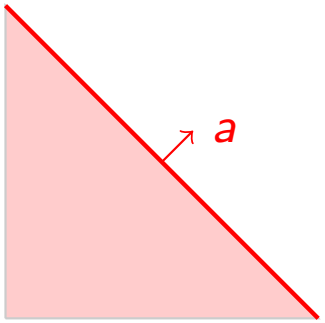
Consider a LP with inequality constraints (easier to visualize).

Definition 5

- $H = \{\underline{x} \in \mathbb{R}^n : \underline{a}^T \underline{x} = b\}$ is a hyperplane
- $H^- = \{\underline{x} \in \mathbb{R}^n : \underline{a}^T \underline{x} \leq b\}$ is an affine half-space (half-plane in $\mathbb{R}^2$).

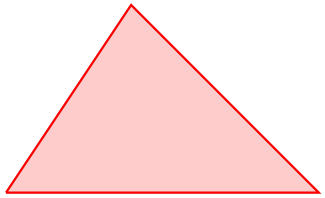
Each inequality constraint ($\underline{a}^T \underline{x} \leq b$) defines an affine half-space in the variable space:

$$H^- = \{\underline{x} \in \mathbb{R}^n : \underline{a}^T \underline{x} \leq b\}, \underline{a} \neq \underline{0}$$



Definition 6

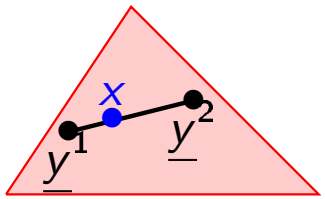
The feasible region X of any LP is a polyhedron P. (Intersection of a finite number of half-spaces.)



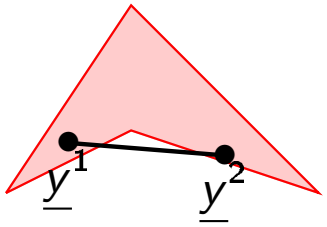
P can be empty or unbounded

Definition 7

A subset $S \subseteq \mathbb{R}^n$ is convex if for each pair $\underline{y}^1, \underline{y}^2 \in S$, S contains the whole segment connecting $\underline{y}^1$ and $\underline{y}^2$.



convex



not convex

Definition 8

The segment defined by $\underline{y}^1, \underline{y}^2 \in S$, defined by all the convex combinations of $\underline{y}^1$ and $\underline{y}^2$, is

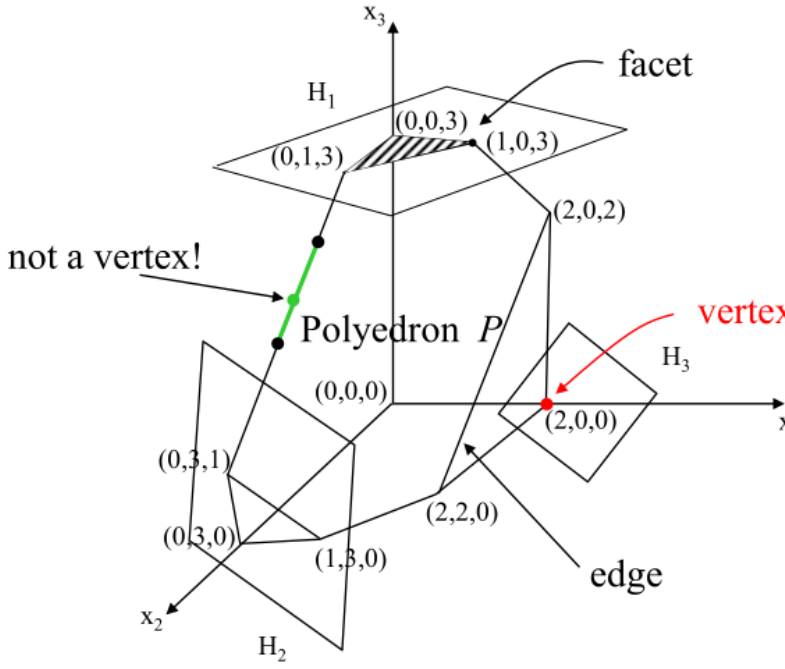
$$[\underline{y}^1, \underline{y}^2] = \{\underline{x} \in \mathbb{R}^n : \underline{x} = \alpha \underline{y}^1 + (1-\alpha)\underline{y}^2 \wedge \alpha \in [0,1]\}$$

Property 1

A polyhedron P is a convex set of $\mathbb{R}^n$.

Indeed:

- any half-space is convex;
- the intersection of a finite number of convex sets is also a convex set.



Definition 9

A vertex of P is a point of P which cannot be expressed as a convex combination of two other distinct points of P.

Algebraically, $\underline{x}$ is a vertex of P if and only if

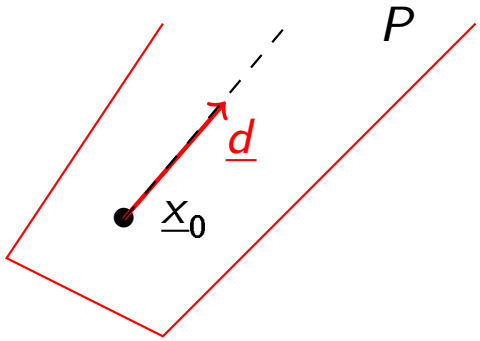
$$\underline{x} = \alpha \underline{y}^1 + (1-\alpha)\underline{y}^2,\ \alpha \in [0,1],\ \underline{y}^1, \underline{y}^2 \in P \Rightarrow \underline{x} = \underline{y}^1 \vee \underline{x} = \underline{y}^2$$

Property 2

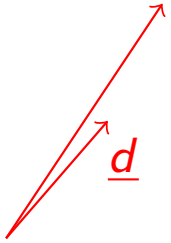
A non-empty polyhedron $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} = \underline{b}, \underline{x} \geq \underline{0}\}$ *(in standard form) or* $P = \{\underline{x} \in \mathbb{R}^n : A\underline{x} \geq \underline{b}, \underline{x} \geq \underline{0}\}$ *(in canonical form) has a finite number* (≥ 1) *of vertices.*

Definition 10

Given a polyhedron P, a vector $\underline{d} \in \mathbb{R}^n$ with $\underline{d} \neq 0$ is an unbounded feasible direction of P if, for every point $\underline{x}_0 \in P$, the "ray" $\{\underline{x} \in \mathbb{R}^n : \underline{x} = \underline{x}_0 + \lambda \underline{d}, \lambda \geq 0\}$ is contained in P.



unbounded feasible directions of P

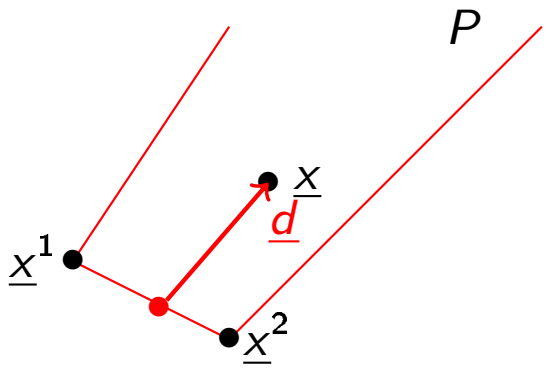


Theorem 11 (Representation of polyhedra – Weyl-Minkowski)

Every point $\underline{x}$ of a polyhedron P can be expressed as a convex combination of its vertices $\underline{x}^1, \ldots, \underline{x}^k$ plus (if needed) an unbounded feasible direction $\underline{d}$ of P:

$$\underline{x} = \alpha_1 \underline{x}^1 + \cdots + \alpha_k \underline{x}^k + \underline{d}$$

where the multipliers $\alpha_i \geq 0$ satisfy $\alpha_1 + \cdots + \alpha_k = 1$.

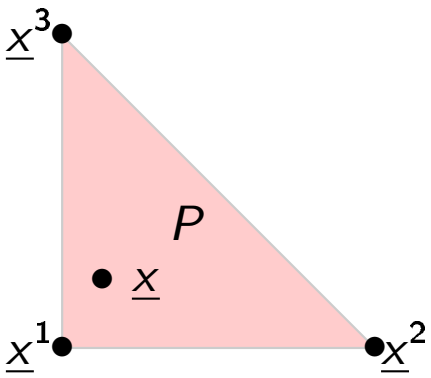


$$\forall \underline{x} \in [\underline{x}^1, \underline{x}^2], \underline{d} = \underline{0}$$

Definition 12

A polytope is a bounded polyhedron, that is, it has the only unbounded feasible direction $\underline{d} = \underline{0}$.

Consequence Every point $\underline{x}$ of a polytope P can be expressed as a convex combination of its vertices.



Then, $\underline{x} = \alpha_1 \underline{x}^1 + \alpha_2 \underline{x}^2 + \alpha_3 \underline{x}^3$, with $\alpha_i \geq 0$ and $\alpha_1 + \alpha_2 + \alpha_3 = 1$ ($\underline{d} = \underline{0}$).

Theorem 13 (Fundamental theorem of Linear Programming)

Consider a LP $\min\{\underline{c}^T\underline{x} : \underline{x} \in P\}$, *where* $P \subseteq \mathbb{R}^n$ *is a non-empty polyhedron (in standard or canonical form). Then either there exists (at least) one optimal vertex or the value of the objective function is unbounded below on* P.

Proof.

- Case 1: P has an unbounded feasible direction $\underline{d}$ such that $\underline{c}^T\underline{d} < 0$.

  P is unbounded and the values $z = \underline{c}^T\underline{x}$ tend to $-\infty$ along the direction $\underline{d}$.

- Case 2: P has no unbounded feasible direction $\underline{d}$ such that $\underline{c}^T\underline{d} < 0$, that is, for all of them we have $\underline{c}^T\underline{d} \geq 0$.

  Any point of P can be expressed as:

$$\underline{x} = \sum_{i=1}^{k} \alpha_i \underline{x}^i + \underline{d},$$

  where $\underline{x}^1, \ldots, \underline{x}^k$ are the vertices of P, $\alpha_i \geq 0$ with $\alpha_1 + \cdots + \alpha_k = 1$, and $\underline{d} = \underline{0}$, or $\underline{d}$ is an unbounded feasible direction.

...

□

Proof.

- Case 2 (cont.): For any $\underline{x} \in P$, we have $\underline{d} = \underline{0}$ or $\underline{c}^T \underline{d} \geq 0$ and hence

$$\underline{c}^T \underline{x} = \underline{c}^T \left(\sum_{i=1}^{k} \alpha_i \underline{x}^i + \underline{d} \right) = \sum_{i=1}^{k} \alpha_i \underline{c}^T \underline{x}^i + \underline{c}^T \underline{d} \geq \min_{i=1,\ldots,k} \{\underline{c}^T \underline{x}^i\}$$

since $\alpha_i \geq 0$, for any i, and $\sum_{i=1}^{k} \alpha_i = 1$.

□

Although the variables are fractional, LPs can be viewed as combinatorial problems.

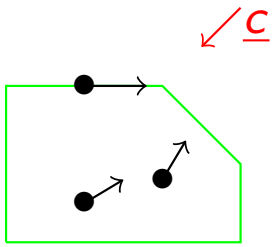
We "only" need to examine the vertices of the polyhedron of the feasible solutions!

The number of vertices is finite but often exponential.

The graphical method is only applicable for $n \leq 3$.

Geometrically

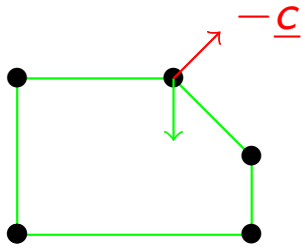
An interior point $\underline{x} \in P$ cannot be an optimal solution:



$\underline{c}$ = direction of fastest increase in z (constant gradient)

$\exists$ always an improving direction

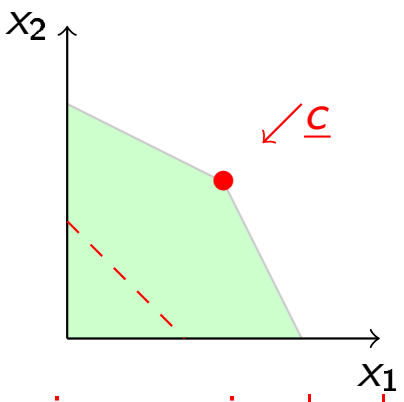
In an optimal vertex all feasible directions (for a sufficiently small step) are "worsening" directions:



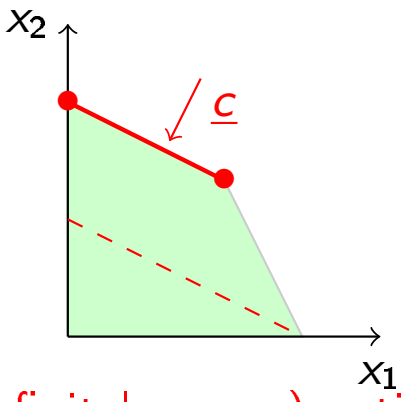
• vertices
improving directions
feasible directions

3.2.3 Four types of Linear Programs

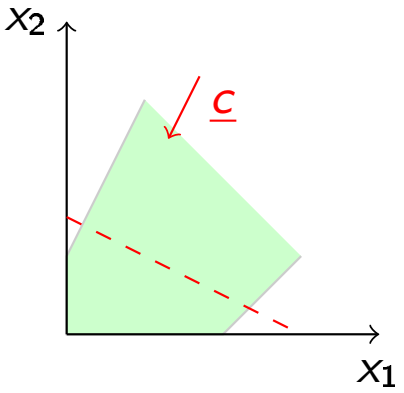
Note: Since $\min \underline{c}^T \underline{x}$, better solutions are found by moving along $-\underline{c}$.



A unique optimal solution

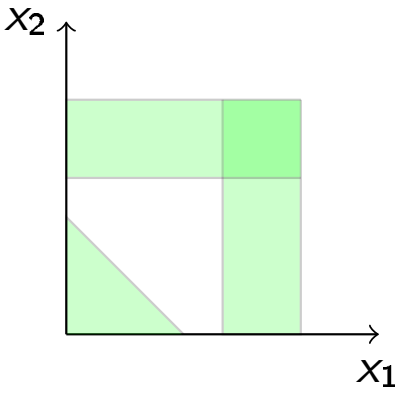


Multiple (infinitely many) optimal solutions



Unbounded polyhedron and unlimited objective function value

Unbounded LP



Empty polyhedron (no feasible solution)

Infeasible LP